Type: III

(i) $\int e^x\,dx = e^x + c$

(ii) $\int e^{ax+b}dx = \frac{1}{a}e^{ax+b} + c$

(iii) $\int a^x\,dx = \frac{1}{\log a}a^x + c,\ a>0$ and $a \neq 1$

(iv) $\int a^{mx+n}dx = \frac{1}{m\log a}a^{mx+n} + c,\ a>0$ and $a \neq 1$

Example 2.16

Evaluate $\int 3^{2x+3}\,dx$

Solution:

$$\int 3^{2x+3}\,dx = \int 3^{2x}\cdot 3^3\,dx$$
$$= 3^3\int 3^{2x}\,dx$$
$$= 27\frac{3^{2x}}{2\log 3} + c$$

DO YOU KNOW?

$$\int ma^{mx+n}\,dx = \int a^{mx+n}\,d(mx+n)$$
$$= \frac{1}{\log a}a^{mx+n} + c,\quad a>0 \text{ and } a \neq 1$$

Example 2.17

Evaluate $\int \frac{e^x+7}{e^x}dx$

Solution:

$$\int \frac{e^x+7}{e^x}dx = \int\left(1+7e^{-x}\right)dx$$
$$= x - 7e^{-x} + c$$

DO YOU KNOW?

$$\int (2ax+b)e^{ax^2+bx+c}\,dx$$
$$= \int e^{ax^2+bx+c}\,d\left(ax^2+bx+c\right)$$
$$= e^{ax^2+bx+c} + k$$

Example 2.18

Evaluate $\int \frac{5+5e^{2x}}{e^x+e^{-x}}dx$

Solution:

$$\int \frac{5+5e^{2x}}{e^x+e^{-x}}dx = 5\int \frac{e^x\left(e^{-x}+e^x\right)}{e^x+e^{-x}}dx$$
$$= 5\int e^x dx$$
$$= 5e^x + c$$

Example 2.19

Evaluate $\int\left(e^x + \frac{1}{e^x}\right)^2 dx$

Solution:

$$\int\left(e^x+\frac{1}{e^x}\right)^2 dx = \int\left(e^{2x}+\frac{1}{e^{2x}}+2\right)dx$$
$$= \int\left(e^{2x}+e^{-2x}+2\right)dx$$
$$= \frac{e^{2x}}{2} - \frac{e^{-2x}}{2} + 2x + c$$

Exercise 2.3

Integrate the following with respect to x.

1. $e^{x\log a} + e^{a\log a} - e^{n\log x}$
2. $\frac{a^x - e^{x\log b}}{e^{x\log a}\ b^x}$
3. $\left(e^x+1\right)^2 e^x$
4. $\frac{e^{3x}-e^{-3x}}{e^x}$
5. $\frac{e^{3x}+e^{5x}}{e^x+e^{-x}}$
6. $\left[1-\frac{1}{x^2}\right]e^{\left(x+\frac{1}{x}\right)}$
7. $\frac{1}{x(\log x)^2}$
8. If $f'(x) = e^x$ and $f(0) = 2$, then find $f(x)$

Type: IV

(i) $\int \sin x\,dx = -\cos x + c$

(ii) $\int \sin(ax+b)\,dx = -\frac{1}{a}\cos\left(ax+b\right) + c$

(iii) $\int \cos x\,dx = \sin x + c$

(iv) $\int \cos(ax+b)\,dx = \frac{1}{a}\sin(ax+b) + c$

(v) $\int \sec^2 x\,dx = \tan x + c$

(vi) $\int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b) + c$

(vi) $\int \operatorname{cosec}^2 x\,dx = -\cot x + c$

(viii) $\int \operatorname{cosec}^2\left(ax+b\right)dx = -\frac{1}{a}\cot\left(ax+b\right) + c$

Example 2.20

Evaluate $\int\left(2\sin x - 5\cos x\right)dx$

Solution:

$$\int\left(2\sin x - 5\cos x\right)dx = 2\int \sin x\,dx - 5\int \cos x\,dx$$
$$= -2\cos x - 5\sin x + c$$

XII Std - Business Maths & Stat EM Chapter 2.indd 32 2/28/2020 5:38:12 PM